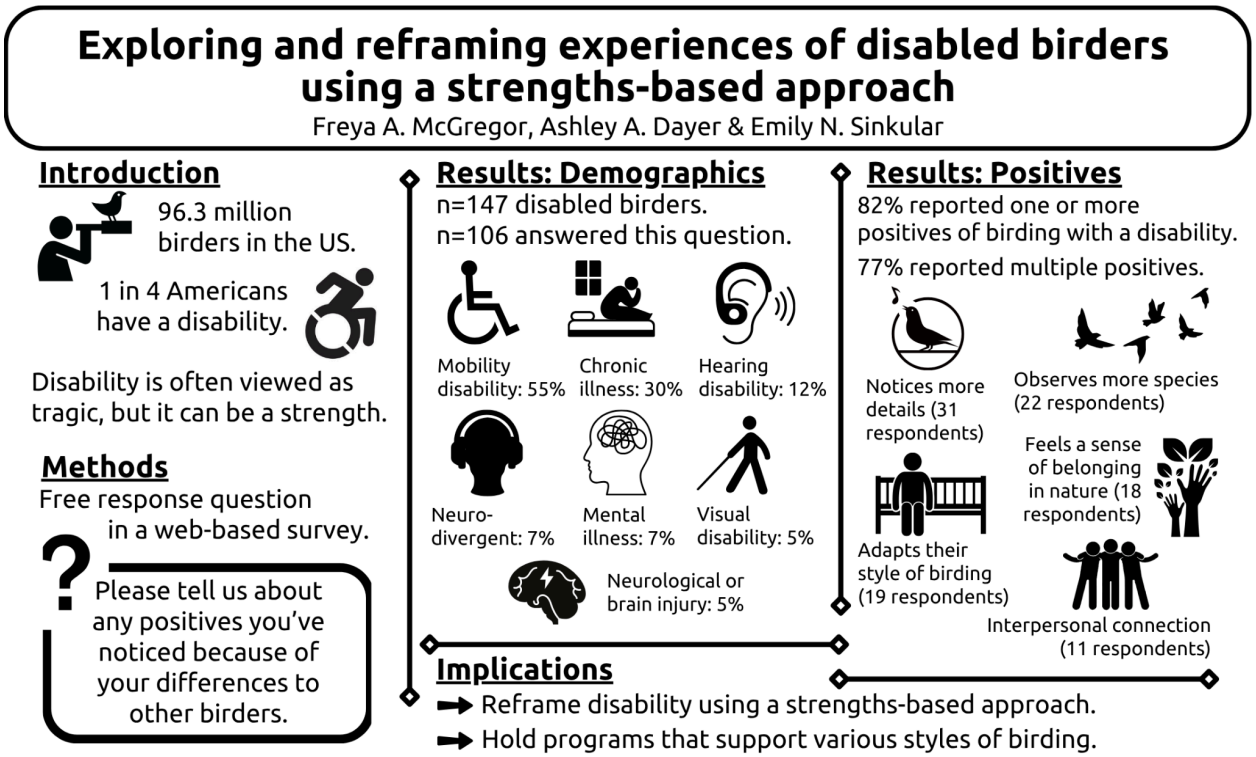
“I appreciate birding more”: Exploring and reframing experiences of disabled birders using a strengths-based approach

Freya A. McGregor[a], Ashley A. Dayer[a] & Emily N. Sinkular[a].

[a] *Department of Fish & Wildlife Conservation, College of Natural Resources and the Environment, Virginia Polytechnic and State University, Cheatham Hall, 310 West Campus Drive (MC0321), Blacksburg VA 24061, USA. Corresponding author: freyamcgregor@vt.edu*

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Abstract

While being disabled is often considered a deficit that creates only problems, in reality every human has interests, values, skills, knowledge, and external supports that can act as strengths when participating in recreation. Indeed, many disabled birders anecdotally report experiencing positives of birding with a disability, and the inherent modifiability of birding may allow disabled birders to experience different ways of engaging with this activity. We conducted a survey of disabled birders (n=147), then used inductive coding to analyze the responses to an open-ended question (n=106) about the positives of

birding they experience. These responses included themes of slowing down, identifying more birds, noticing more details of birds, and adapting their style of birding to meet their access needs. We also used inductive coding to identify the various styles of birding respondents engaged with. This reconsidering of disability using a strengths-based approach creates opportunities for disabled birders and bird-related programming to focus on aspects of birding that may be more in line with disabled birders' interests, experiences and strengths. It also creates a new avenue for nature organizations to engage with up to a quarter of the population who may feel excluded due to disability. This study contributes to the literature by exploring a rarely studied topic of disabled birders' experiences using a strengths-based approach to studying disability in outdoor recreation.

Keywords
Birding, birdwatching, disability, disabled, inclusion, strengths.

Highlights

- People with a variety of disabilities engage in birding (or bird watching).
- Birders with disabilities report a variety of positives of birding differently.
- Disabled birders report noticing more details, and observing more species of birds.
- Reframing "deficits" as strengths in birding may push against ableism.
- Using a strengths-based approach may support inclusion of birders with disabilities.

Management implications
Using a strengths-based approach to birding with a disability, land, program and recreation managers can support people with disabilities to engage in birding, such as by providing opportunities for stationary birding and incorporating birding by ear skills into birding programs. Designing and building more accessible birding locations, such as by incorporating more benches along trails, building observation blinds or platforms with ramps to the entrance rather than only steps, and using firm and stable trail surfaces, would likely support people with disabilities in engaging in birding. Reframing disability as a strength in birding may support more people with disabilities to engage in outdoor recreation and help push back against the pervasive negative perception of disability.

1. Introduction

Birding, enjoying wild birds by watching, listening or otherwise noticing (McGregor, 2021), is a popular activity throughout the world (e.g. Ma et al., 2012; Randler et al., 2022; Randler & Dutour, 2025). The U.S. Fish and Wildlife Service (2023) reports that in

2022, 96.3 million Americans aged 16 years or older engaged in birding (which includes closely observing, photographing, and/or feeding birds); of these, 95% observed birds around the home. While estimates for birding participation do not exist in other countries, we know that in the UK 48% of households do so (Davis et al., 2009), and the estimated proportion of households in Australia is 36-48% (Ishigame & Baxter, 2007). Birders in Austria, Germany and Switzerland are most strongly motivated to participate in birding by the opportunity to experience nature, their own fascination with birds, their enjoyment of being alone, and to improve their birding-related skills (Randler, 2023), while people who feed birds in New Zealand were most strongly motivated by the pleasure they derived from doing so (Galbraith et al., 2014). Birders' overall health and wellness also improve as a result of birding, with reported benefits including feeling more relaxed (Cox & Gaston, 2016), an increased sense of wellbeing (Hammond et al., 2022), improving learning and memory (Zieris et al., 2023), and feeling a sense of companionship with birds and belonging in nature (Bell, 2019).

Although demographic or identity questions related to disability have not been included in any other studies of birders the authors are aware of (see, for example, U.S. Fish and Wildlife Service, 2023; Patton & Fulton, 2021), Sinkular et al. (2024) in their U.S. panel survey found that 39% of birders reported experiencing *somewhat*, *quite a bit*, or *a great deal* of accessibility challenges as a result of a disability when engaging with feeding or observing wild birds. Birding is inherently flexible, as it can be done from home, on trips away from home, by ear, while stationary, on foot, by car, etc (McGregor, 2021). This flexibility allows this activity to be modified in a myriad of ways to support birders with accessibility challenges, and to take into account their activity restrictions, energy or fatigue levels, or other factors resulting from disabilities, injuries or illnesses.

One in 4 Americans (61 million people) live with a disability (Centers for Disease Control and Prevention, 2023), and globally 1 in 6 (1.3 billion people) are disabled (World Health Organization, 2023). Disability can be caused by a variety of factors, from chronic illnesses (such as multiple sclerosis or chronic fatigue), to injuries (including spinal cord injuries or traumatic brain injuries), to diagnosed medical conditions (including cerebral palsy or macular degeneration) and mental health conditions (including bipolar disorder or obsessive compulsive disorder), and can be temporary, episodic, or permanent (Centers for Disease Control and Prevention, 2024). Different types of disabilities affect daily life in different ways (Centers for Disease Control and Prevention, 2024) and may require different modifications or accommodations to ensure access needs are met (Reinholz, & Ridgway, 2021).

Globally, the overall estimate of neurodivergent prevalence is 15-20% (Doyle, 2020). "Neurodivergent" is a community-generated term, rather than a diagnosis, which acknowledges that information is received and processed by various brains in ways that are different from the "typical" brain (Walker, 2021). Many people who are autistic, have attention deficit hyperactivity disorder and/or other diagnoses identify with this term.

Historically, many educators and healthcare providers have attempted to reduce or remove neurodivergent traits from individuals' behavioral repertoire with little success (Dallman et al., 2022). This may be done, for example, by pressuring autistic people to mask (hiding or suppressing autistic behaviors; Evans et al., 2023) which can result in autistic burnout, depression and anxiety (Evans et al., 2023). Meanwhile, sensory processing sensitivity – a common experience of neurodivergent people (e.g, Baron-Cohen et al., 2009) – can be distressing and disabling, especially in noisy environments (e.g., Aron, 2004).

Despite the United Nations Convention on the Rights of Persons with Disabilities (2006) commitment to ensuring people with disabilities enjoy all human rights and freedoms, disability has long been stigmatized and viewed by many cultures as a negative, with disabled people seen as a burden and something to be pitied (e.g., Gill et al., 2016; Mitter et al., 2019; World Health Organization, 2023). The persuasive cultural stigma associated with disability can become internalized and impact someone's self-esteem (Paterson et al., 2011), their self-reported quality of life (Monden et al., 2021), their employment opportunities (Carolan, 2023), and their overall mental health (Cadden, el al., 2018). A recent analysis found that "birders often expressed feelings of shame, embarrassment, or self-consciousness associated with their disability" (Sinkular et al, 2024, p. 10). Shame is "an intensely painful feeling or experience of believing we are flawed and therefore unworthy of acceptance and belonging" (Brown, 2006, p. 45), and can result in decreased feelings of self-worth (Clapton et al., 2017), making it difficult to connect a positive identity to being disabled (Jóhannsdóttir et al, 2021), and contributing to poor mental health outcomes (Brown & Turner, 2010; Clapton et al., 2017). However, the concept of the disability paradox describes the phenomenon that many people with disabilities, despite the stereotype of disability being tragic, report a high level of life satisfaction (Albrecht & Devlieger, 1999). Being integrated into social groups through mutually-enjoyed activities, such as being an active member of a bird club, can reduce negative stigma associated with having a disability (Taub, et al., 1999), and may invite all people to feel more comfortable being their own authentic selves – which may include identifying as disabled (Dunn & Burcaw, 2013).

A strengths-based approach to disability focuses on what a person is able to do, rather than what they cannot do. Strength-based approaches identify what factors in someone's environment are supportive, and emphasize growth and self-empowerment, aligning their functional capacity with the environment they are in (e.g. Heyne & Anderson, 2012; Niemiec & Tomasulo, 2023). Strengths include someone's abilities, skills, interests, values, character traits, and their resources and other supports (Heyne & Anderson, 2012; Niemiec & Tomasulo, 2023), as well as external factors such as inclusive communities, community resources, and family and friend support (Heyne & Anderson, 2012). Rather than focusing on the negative experiences – to which our brains default (e.g. Rozin & Royzman, 2001; Niemiec & Tomasulo, 2023) – or trying to

change one's character or physical weaknesses, amplifying and applying one's strengths often yields greater benefits. A wealth of studies (see Table 1 for examples) supports the idea that focusing on our innate character traits, what we are good at doing, and the resources and supports we have available to us supports our wellbeing (e.g. Biggs & Carter, 2015; Kirchner et al., 2016).

Table 1: Select benefits of using a strengths-based approach for people with and without disabilities

Benefit	*Reference*
Increased motivation and confidence in the workplace	Hiemstra & Van Yperen, 2015
Reducing frustration in the workplace	Gradito Dubord & Forest, 2022
Increasing motivation for learning in the classroom	Hiemstra & Van Yperen, 2015
Improved learning outcomes	Callingham, 2013
Improved self-esteem while learning a second language in university settings	Narafshan & Noori, 2018
Improved mental health for cancer patients	Craske, et al., 2019
Improved mental health for people living with depression	Cheavens et al., 2012
Increased resilience over a three month period for people with acquired physical disabilities	Bu & Duan, 2021
Individual, family, community and environmental/land strengths fostered increased health and wellbeing for Indigenous youth and adults around the world	Roundtree & Smith, 2016

Indeed, using this approach may help change the harmful and stigmatizing view many hold about disabled people (Hayden & Prince, 2020; Niemiec & Tomasulo, 2023). In healthcare fields, such as occupational therapy, providers are moving towards a strengths-based approach, promoting and building upon a person's strengths and talents rather than focusing on trying to remediate limitations or deficits (Dunn, 2017). A strengths-based approach to disability draws on the social model of disability (which views the environment as the factor that disables someone, rather than the person being defective; Oliver, 2013; Marks, 1996), the identity model of disability (which celebrates belonging within the disability community; Retief & Letšosa, 2018), and the functional model of disability (which is concerned with what someone can do, rather

than what they cannot do; Smart, 2001). This approach is not about ignoring the very real challenges many people live with and forcing a falsely positive attitude, but on amplifying and nurturing skills and traits that someone is naturally good at, instead of focusing on trying to turn someone into someone they are not.

Applying a strengths-based approach, the purpose of this research is 1) to determine if strengths-based thinking resonates with birders with disabilities; and 2) to identify any positives or benefits that disabled birders experience. As far as we are aware, whether a strengths-based approach to birding or other forms of outdoor recreation resonates with recreationists has never been studied, although Heyne and Anderson (2012) and Anderson and Hayne (2012; 2013) have written about how strengths-based approaches to therapeutic recreation could be utilized.

2. Methods

To investigate any benefits experienced by, or positives of, birding with disabilities, the first author conducted a survey of disabled birders between October and December 2020, prior to working at her current institution. Questions were developed through anecdotal evidence from other disabled birders and informed by the first author's professional background as an occupational therapist, personal experiences as a disabled birder, and skilled activity analysis and observations of birding locations and the social environments of various birding communities.

The survey, hosted in Google Forms, was composed of 15 questions. A combination of purposive and snowball sampling was used to distribute the survey (Taylor, 2017). A link to the survey was shared after virtual events related to disability and birding, via the first author's social media accounts, and emailed directly to known disabled birders inviting them to complete it. Participation in the survey was voluntary and anonymous.

The first author began collaborating with the coauthors in 2021 when the coauthors received a de-identified dataset for analysis from her. The Virginia Tech Institutional Review Board (IRB #21-1074) deemed this analysis and manuscript writing "not research involving human subjects". The first author began working at Virginia Tech in 2022.

The first author began working at Virginia Tech in 2022 and the coauthors' received a de-identified dataset for analysis, which was deemed "not research involving human subjects" by their university and was conducted with approval from the Virginia Tech Institutional Review Board (IRB #21-1074).

To address the research objective, the first and third authors qualitatively coded responses to the following question: "Some people notice positives to having different abilities and experiences than others. For example, if they have to walk more slowly than others, they may see birds that others have rushed past and missed. Please tell us about any positives you've noticed because of your differences to other birders."

Inductive coding was completed, which uses no predetermined codes and allows the concepts and the codes to emerge from the data (Elliott, 2018). Codes used and their explanations are shown in Tables 2, 3 and 4. Because themes were allowed to emerge from the responses, it is possible that some of them would be experienced by nondisabled birders too.

To increase inter-rater reliability, the second author reviewed the results after the initial coding, with all three authors agreeing on the final coding decisions after discussion whenever discrepancies occurred (Belur et al., 2021). A code was counted only once for each respondent, but there was no limit to the number of codes one response might include. Quotes from survey respondents are shared in this paper alongside information about their disability to provide context for why their strength may be related to their disability.

This research was conducted by three white, cisgender women who speak English as their first language and live in the United States. One is an Australian occupational therapist, and two are American conservation social scientists. All three are birders with their own lived experience of disability, including mobility-related disabilities, mental health conditions, chronic illness, cancer and/or neurodivergence, and are caregivers to others with one or more of these lived experiences. Our birder, disability and caregiver identities may have impacted the lens through which we engaged with the data; however, without these identities, we feel it likely we would not have developed or explored this research question.

2.1 Demographics

The survey asked participants a free-response question: “The primary disability, illness, injury, medical condition or other health concern that impacts my ability to go birding is. (It's OK not to answer if you'd rather not.)” And in the following question, “Other disabilities, illnesses, injuries, medical conditions or other health concerns that impact my ability to go birding are. (It's OK not to answer if you'd rather not).” Using a diagnostic and functional approach the responses from these questions were coded into broad disability categories by the first author (Table 2), and the number of respondents who live with each disability category is shared in Figure 1.

Table 2: Codebook used for grouping respondents into broad disability categories. Note that the disabilities, illness or injuries mentioned in the right column are only those mentioned specifically by respondents, and do not include other diagnoses that could fall into the disability categories more broadly.

Disability category	*Diagnoses, conditions or access challenges included in this category*
Mobility disability	People who have spinal cord injuries and multiple sclerosis, who use mobility devices such as walking frames, who have polio or

		muscular dystrophy, who reported having ‘bad knees’ or arthritis, and who have balance issues or difficulty walking or standing.
	Spinal cord injury	People who stated they have a spinal cord injury.
	Multiple sclerosis	People who stated they live with multiple sclerosis (MS).
Chronic illness		People who live with chronic fatigue, fibromyalgia, Lyme disease, lupus, Ehlers-Danlos syndrome, irritable bowel syndrome and asthma.
Deaf, deaf or hard of hearing*		People who primarily use sign language to communicate, or who have mild to severe hearing loss.
Neurodivergent or learning disability		People who are autistic, have attention deficit hyperactivity disorder (ADHD), or have a learning disability.
Mental illness		People living with depression, anxiety, and/or post traumatic stress disorder (PTSD).
Visual disability		People who are totally blind, or who have permanently reduced vision that can not be corrected with glasses, contact lenses or surgery, in one or both eyes.
Neurological condition or brain injury		People who have had a stroke or a traumatic brain injury (TBI), or who have epilepsy.
Not stated		No information was provided about their type of disability.

*Referring to the Deaf community with a capital “D” respects their culture, community, and use of a separate language (sign language); meanwhile, lower case “deaf” is used by many people who have hearing loss but do not share the same culture as those in the Deaf community (DiMarco, 2022). Other people with hearing loss may identify as “hard of hearing”. As such, when referring to this entire demographic, we use the full phrase of “Deaf, deaf and hard of hearing”.

2.2 Positives of being a disabled birder

We coded responses related to the positives birders with disabilities noticed (see Table 3 for the codebook used, and Figure 2 for the number of responses in each category). If a respondent mentioned more than one positive, their response was coded into each relevant category.

Table 3: Codebook used for the qualitative coding of survey responses about the positives of birding differently than others.

Code used	*Explanation*
Notices more details	Notices more details including colors, behaviors, habitat, nests etc. Used if they stated, “I see great things” or similar.
Notices more birds	Observes a higher species count on a bird outing, e.g., “more birds”, “more species”. May also include observing other wildlife.
Adapting birding style/s	Clearly indicated they have modified how they go birding to meet their access needs as a result of an acquired disability or progressive disease.
Belonging in nature	Experienced a sense of belonging or community with nature. This code was used if they stated that birds “come to me”.
Interpersonal connection	Experienced a sense of community with fellow birders. Responses must explicitly relate to making a connection with others. If a response simply mentions other birders, that is not automatically “connecting” with them.
Knowledge sharing opportunities	Contributed to or benefitted from individual or collective knowledge about birds, e.g., through community interaction or by participating in participatory science projects.
Gratitude for birding	Felt thankful or appreciated birds and/or their own birding experiences. Only used if this was explicitly mentioned; not used if it was a general idea.
Nature-related creativity	Participated in creative activities inspired by birds or nature, such as photography, journaling or other arts or crafts.
Adaptive equipment as a facilitator	Any object used to enable a person to participate in birding beyond standard binoculars or spotting scopes, field guides or identification apps, e.g., bringing their own chair to use on the trail, or using a power wheelchair. This is considered a strength because using adaptive equipment requires the person to acknowledge that the equipment would support

	them, and then actively choose to use it.
Having time available	Stated they have enough time available to spend birding. Response must have connected having time available to having a disability.
Flexible expectations	Adjusted their expectations in a day-to-day way for any given experience. This may have been expressed that they went birding with one expectation, and it wasn't met, but they didn't mind.
Opportunity for exploration	Indicated that birding prompts them to visit new places or discover more about one place, e.g., "finding a quiet spot" or "exploring the trail".
Health/ wellness benefits	Explicitly stated they experience physical, mental and/or emotional health and wellness benefits from birding. This may include feeling refreshed, relaxed or energized after birding, and escaping from reality for a time.
Negative reported	Response indicated a negative or less desirable outcome.
No positives identified	"None", "N/A" or similar.

2.3 Styles of birding used

As analysis was conducted, it became clear that many of the strengths respondents shared were directly related to the inherent modifiability of birding – the fact that birding can be adapted in various ways was what allowed these respondents to use their strengths to participate in this activity. Inductive coding was completed regarding these 'styles' of birding used by respondents; Table 4 shows the codebook used.

Table 4: Codebook used for the qualitative coding of responses that provided some information about the style of birding used.

Code used	*Explanation*
Stationary birding	Indicated they do some, or all, of their birding from one position. Note this may not always be seated.
Birding with others	Indicated they engage with other people while birding. This may include a friend or partner, on an organized bird outing, or with another random birder they interact with.
Mindful birding	Indicated they are fully present and attending to the birds, rather than, for example, seeking more species or focusing on

	identifying birds. “Patient” is considered mindful.
Birding by ear	Listened to bird sounds, with or without an intent to identify birds by sound. May use in conjunction with birding visually, or may bird exclusively by ear.
Birding along a trail	Explicitly stated they use trails while birding.
Birding solo	Explicitly stated they go birding alone.
Backyard/local birding	Indicated they bird around the home. Includes references to birding in their backyard, neighborhood or at nearby parks.
Bird photography	Explicitly stated they take photos of birds.
Car birding	Stated the engage in birding from a vehicle. This includes golf carts.
Bird sound recording	Explicitly stated they use equipment to record bird sounds.

2.4 Limitations

It is possible that the phrasing of the survey question, including the example that was provided to clarify the meaning of the question, may have made responses vulnerable to suggestibility bias (Erčulj & Šulc, 2025). Suggestibility bias may occur when respondents have been influenced to respond in a certain way, such as by a leading question (Erčulj & Šulc, 2025). The two most common codes (*notices more details* and *notices more birds*) are related to the example we provided in the question, suggesting that respondents may have experienced suggestibility bias. However, participants did still provide a breadth of responses.

This question was included in a larger survey that was solely related to being a disabled birder, however this question did not specifically mention the words “disabled” or “disability”, and instead used “different abilities”. Additionally, we only had short answers to analyze and could not seek further clarification from respondents, making it possible that some of the responses provided did not relate specifically to the respondents’ disability(ies), but rather to their experiences of birding in general. Disability is complex, and often absorbed as an identity (e.g. Nario-Redmond, et al., 2013; Santuzzi & Waltz, 2016; Bogart et al., 2017), so unless it was stated explicitly, we did not feel it would be appropriate to attempt to determine on behalf of respondents which responses were related to their disability(ies), and which (if any) were not.

Although Google Forms states on their website (Google, 2023) that this software is screen reader accessible, the first author was informed by potential respondents after the survey closed that they had difficulty accessing the survey using their screen reader

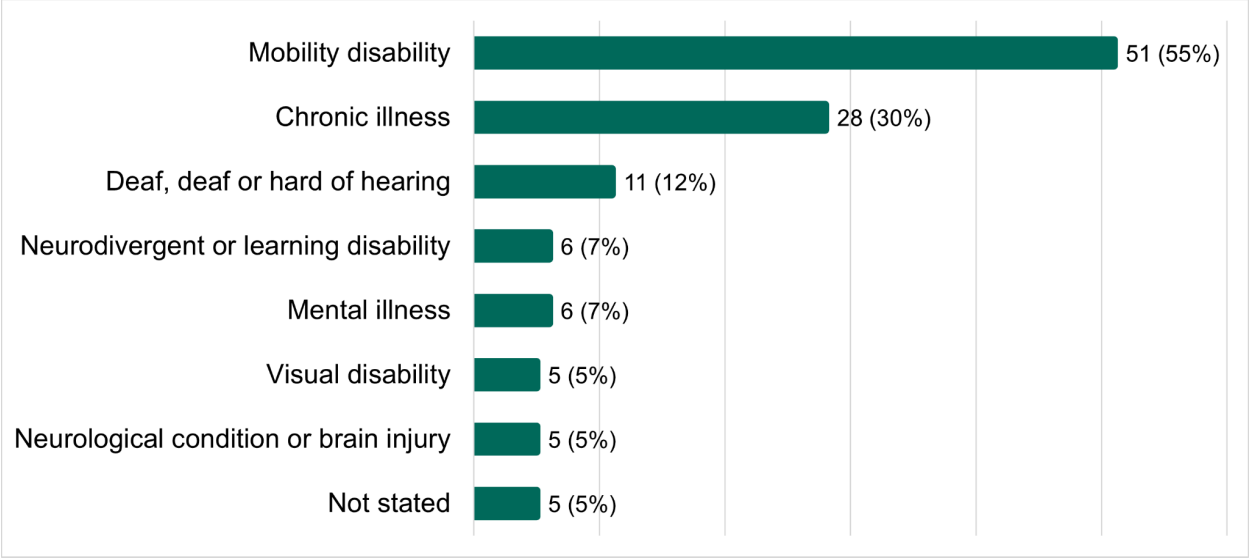
software. This may have impacted the sampling by excluding respondents who rely on this technology (Oh et al., 2021), including, potentially, people who are blind or have low vision or dyslexia. Survey developers should ensure that their surveys follow accessibility guidelines, including that they are screen reader accessible, and any software that is developed to collect data should be designed with these features built-in from the outset. Additionally, end users who rely on screen reader software should be incorporated into the design and user testing phases before any survey is launched (Oh et al., 2021).

3. Results

3.1 Demographics

Disability categories of survey respondents are shown in Figure 1. Due to the level of detail provided in some of the open-ended responses, the category of *mobility disability* was further broken down into *spinal cord injuries* (11 respondents; 12% of respondents), and *multiple sclerosis* (6 respondents; 7% of respondents).

Figure 1: Results of qualitative coding that grouped respondents (n=106) into broad disability categories, in order of frequency. Percentages rounded to the nearest whole number. Note that 31 respondents (34%) reported more than one type of disability, resulting in the percentage of responses not adding to 100%.



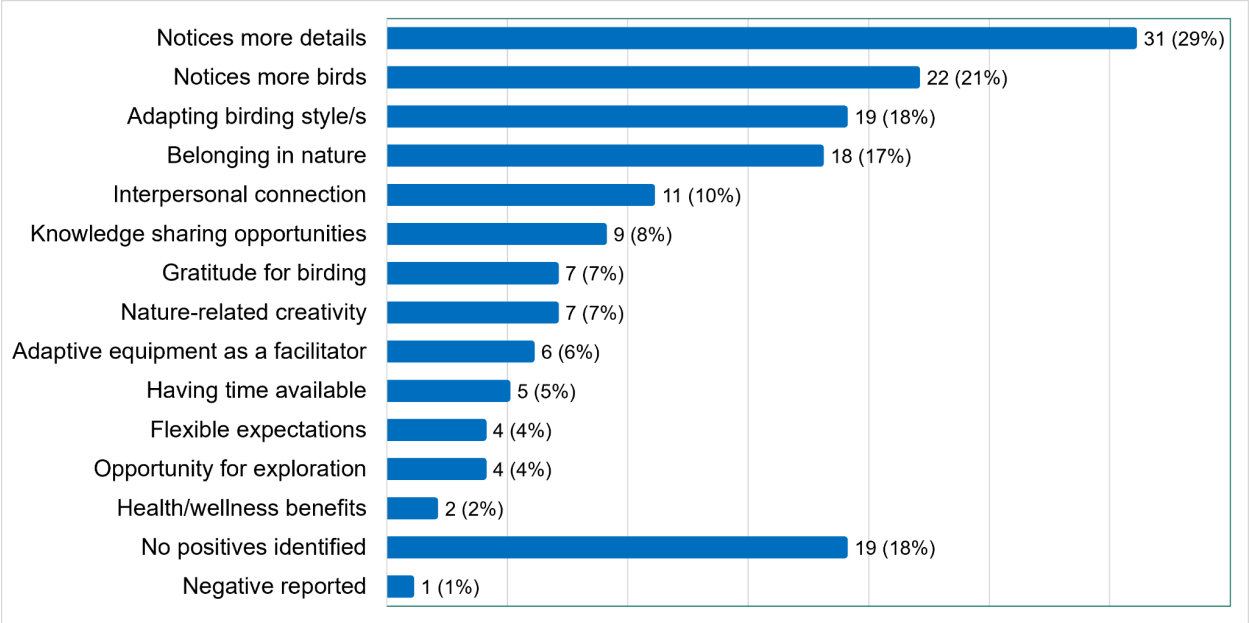
3.2 Positives of birding differently to others

Of the 147 disabled birders who completed the survey, 106 (59%) answered the question, above, about noticing any positives to being a disabled birder. Coding uncovered thirteen distinct positives to birding differently to others (Figure 2; detailed results by code follow).

Of the participants who answered this question, 87 (82% of the sample) shared one or more positives. Eighty-two responses (77% of participants who responded to this

question) had more than one code assigned, indicating they experienced various positives. Nineteen respondents (18%) answered this question with “None”, “N/A”, “I cannot think of any positives” or similar. Two people responded to this question about themselves *and* on behalf of a loved one with a disability; the remaining respondents responded only about their own personal experiences.

Figure 2: Results of qualitative coding of survey responses about the positives of birding differently than others, in order of frequency. Percentages were determined based on the number of people who answered the question (n=106), rather than the total number of respondents, and have been rounded to the nearest whole number.



3.2.1 Notices more details

For some birders, having a long list of bird species is considered more valuable, but for others, observing behaviors (“see[ing] great things”) and more details is more rewarding. Thirty-one respondents reported they *notice more details* of birds and the environment as a result of being a disabled birder. An autistic birder shared, “My hearing is extremely sensitive. I can pick up and identify birds by sounds most people don't seem to hear. I'm especially good at picking up hummingbird buzzing.” Another birder with a neurological condition responded, “I can see great distances and I can see a depth of colour [*sic*] and tonal variation making wildlife very visible.”

For some birders with sensory loss, they tuned in to more details through their other senses. One birder with hearing loss found that other environmental clues were helpful: “Because I don't hear well, I've become a much better visual birder. I am more prone to noticing visual cues of a bird's presence - habitat, nest types, material they might use, food sources, etc than some of my birding friends with good hearing.”

For others, *noticing more details* was a side benefit of being left behind by a group; one, with a mobility disability, shared “You can sometimes see great things while

sitting still waiting on a group that may have gone a ways on a non-accessible path/trail."

3.2.2 Notices more birds

Twenty-two respondents reported they notice a broader variety of bird species or other wildlife. One, with a chronic illness and mental illnesses, remarked, "Having to stop and take breaks means I spend time settled down and catch more birds." Another, who lives with a mobility disability post-spinal cord injury, shared, "There's no doubt that I have seen more birds since my injury than before, due to being forced to slow down."

Another respondent connected *noticing more birds* to having hearing loss:

> "Often non-hearing challenged birders will bird without actually paying attention to what is going on around them. They are counting on their ears to inform them of a bird's presence. Not being able to hear the birds, I have a tendency to see more species because I notice the birds that aren't calling or singing [which is] especially helpful in the fall."

3.2.3 Adapting birding style/s

Nineteen respondents (18%) indicated that they had adapted how they go birding as a result of a permanent injury or chronic illness, or an acquired or progressive disability. Some reported learning to rely on other sensory input due to sensory loss: "When my vision deteriorated, I began to Tune In [*sic*] more to my hearing in order to identify birds, something I'd only considered auxiliary until then. I knew that I was one of the youngest birders I had ever met, but I hadn't quite realized until then that I have pretty good hearing." Others indicated they have had to slow down: "My chronic illness has taught me to be very patient and remain still and quiet for longer periods of time."

3.2.4 Belonging in nature

Feeling a sense of community or belonging with birds or nature was reported by 18 respondents. One, who has chronic illnesses and memory loss, shared, "I am doing most of my birding in my garden. I get to know the shifting of birds during the year very well. When the wren comes, it is winter."

Others connected *stationary birding* with birds coming to them. This idea was included in the code of *belonging in nature* because the birds choosing to appear for these people helped them feel like they were part of nature. One respondent, who lives with mental illness, shared, "I've been able to have flocks of ibis & egrets feel comfortable enough to walk near me, by me to get to their ponds. I am very quiet. Very still. I don't like to take their pic[ture]s when they're close. I'm very in the moment with the experience." Another, who has a mobility disability, reported, "Sometimes when I wait alone in the parking lot, birds seem to forget a human is there and go about their normal behaviors."

3.2.5 Interpersonal connection

Eleven responses, or 10% of participants, conveyed that their interactions with others or a sense of community with fellow birders was a positive they experienced, and was described as tied to various activities. For one respondent with hearing loss, this came from taking on a leadership or mentoring role for others: “Because I grew up with very few POC [People of Color] or female birder role models, I used my leadership skills to help teach the bird class at [university redacted] and actively seek to encourage new/younger birders.”

For other birders, the sense of community came from unplanned interactions with others while out birding, such as for one with a chronic illness: “I get to interact with usually really enjoyable people as I ask for help with an ID [identification of a bird] if my brain fog is acting up.”

Other respondents felt included by the birding community because their disability highlighted other birders’ willingness to respect their access needs, like this respondent who has a visual disability: “Because I have to do most of my birding by ear, folks quiet down and pay attention to sound cues.” And, “Birders have been happy to spot the birds and then move aside so I can see [them from my wheelchair].” Another multiply-disabled birder remarked that they were glad they could serve other birders in a different way: “I move slower. I notice accessibility issues for others.”

3.2.6 Knowledge sharing opportunities

Nine respondents shared that the chance to share their bird-related knowledge is a positive aspect of being a disabled birder. Seven of these birders shared their knowledge with other birders they interacted with in the field, while two appreciated sharing their knowledge with birders who had complementary strengths. One birder who has hearing loss, for example, responded, “I've definitely been able to recognize birds by sight generally more than others and can point out where I see them. Birding with those that can hear really help me to know what types of other birds are around and where to look.”

3.2.7 Gratitude for birding

Seven respondents reported that they felt thankful for birds and/or that they could participate in this activity. One birder with a chronic illness shared, “Being grateful for ANY birds I see, even very common ones, since I can’t get out to see them much.” Another birder with a mobility disability responded, “Just being able to get outside at a park or preserve (and shoot pic[ture]s) is a wonderful blessing.”

3.2.8 Nature-related creativity

Seven respondents indicated that the ability to participate in creative outlets such as photography, journaling, or other art is a positive they experience as a result of being a disabled birder. These activities could be used to supplement observation skills:

“Because my hearing loss prevents me almost entirely from using bird song, I learned how to illustrate to learn bird species and am very good at structure and color.”

3.2.9 Adaptive equipment as a facilitator

Six respondents mentioned using adaptive equipment, including power wheelchairs, chairs with neck rests, and golf carts covered in camouflage material, all of which were related to a mobility disability. We considered using adaptive equipment a strength because by using it the person is able to participate more comfortably, independently, or for a longer period of time. One respondent shared, “My power [wheel]chair has a very long range so I can bird for a very long time without getting fatigued!” Another reported, “Sitting in a chair with a neck rest, I can look up into trees a bit easier than most people standing or walking.”

3.2.10 Having time available

Five respondents shared that the increase in time they have available due to their disability was a positive, as it allowed them more time to go birding. A wheelchair user shared, “Having more time to go birding because other activities aren't possible [and] being able to sit anywhere and look up for long periods of time.” Another respondent with a mobility disability shared, “Because of my lack of mobility, I can spend more time with the local birds and [attending] Audubon [events] online.”

Another respondent with a spinal cord injury had been able to explore creative pursuits because of this additional available time: “Since becoming disabled I have made use of free time by learning to draw and do nature journaling. Things I had always wanted to practice but was too busy for.”

3.2.11 Flexible expectations

Approaching birding experiences on a given day without a rigid plan was something that four respondents shared as a positive of being a disabled birder. For one respondent, flexible expectations were connected to their needs when birding with a group:

> “Before I was diagnosed with MS [multiple sclerosis], I struggled with encroaching limitations which caused me to fall back saying ‘I'll see you guys later.’ I've had some lovely bird encounters as a result that I'm sure would not have occurred with a large group.”

Coming with flexible expectations also helped when the physical accessibility of a birding location didn’t meet the birders’ access needs, such as for this birder with a spinal cord injury: “My go-to adaptation to poor access is to find a reasonably quiet, out-of-the-way spot and erect my camo-tent blind and let the birds come to me.”

3.2.12 Opportunities for exploration

Four respondents shared that they visit new places locally or overseas, or find different locations for birding. One, who has a chronic illness, shared, “I found one of my favourite [*sic*] places to bird specifically BECAUSE it was flat and the trail was paved.”

3.2.13 Health/wellness benefits

Two respondents shared that they experience mental, emotional, and/or physical health and wellness benefits from birding with a disability. One, who lives with a chronic illness, shared, “My disability is what inspired me to start birding, because it is a quiet and slow task. I find them [birds] joyful and can forget about my grief over my illness.”

Another birder, who lives with multiple chronic illnesses, found motivation and rejuvenation from spending time in nature:

> “When I need to sit down and rest while the others go ahead, I do see nature differently. I notice mushrooms, birds, nature prints, sounds, deer, birds, etc. I find that rest in the small things of nature is beautiful and helps with my spirits and then I can get up and go again.”

3.2.14 Negatives of being a disabled birder

One response reported both a positive and a negative; this was the only negative reported: “I am able to notice details others don't because of my mental differences [ADHD] which can also be a downside as sometime[s] I get caught up in a small detail, missing another bird.” Meanwhile, 18% of respondents (19 respondents) did not identify any positives to birding with a disability, with responses such as “N/A” or “None” most commonly shared by this group.

3.2.15 Various positives identified by many respondents

Eighty-two responses (77% of birders who answered this question, and 92% of those who identified positives) referenced two or more positives to having different birding experiences than others as a result of their disability. These were coded into each of the applicable categories above. Such responses included this one, with relevant codes identified in square brackets, from a birder with a chronic illness and a neurological condition:

> “OMG because I bird mostly from my van I get to know the birds in the area I'm in very well - the different groups of black capped chickadees that live around the feeding stations, the pairs of Canada jays, this year I got to meet and watch one family of Eastern Phoebes while they nested, hunted and raised their kids [*notices more details*]. Lots of twitchy tails, lol. And you get to listen, really listen, [*mindful birding; listening experiences*] and I've learned more bird songs and calls. I find it makes me feel closer to the birds - I say good morning when I arrive and often say thank you [*gratitude for birding*] for the beautiful pictures they allow me to take [*belonging in nature*].”

The most common combination was *slowing down* and *visual experiences*, with 13 responses (11%) being coded into both categories. *Slowing down* was also correlated with *noticing more birds,* with 12 responses being coded into both, such as this one,

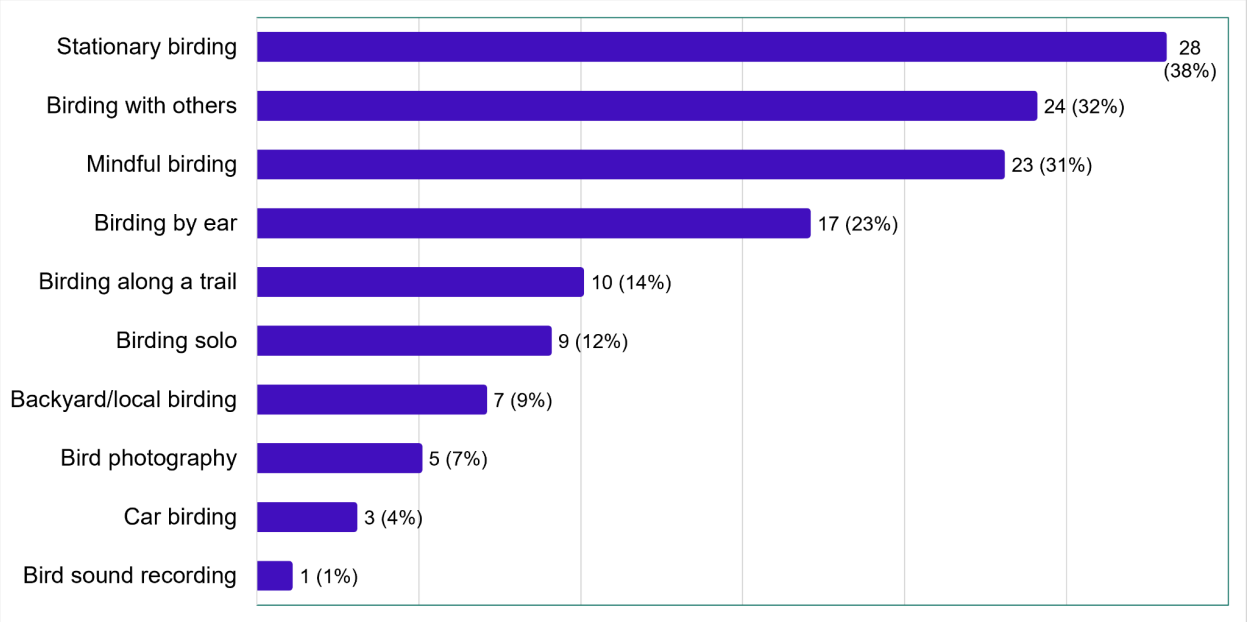
from a birder with a neurological condition, “Because I do move slowly I do notice birds and or other wildlife more often.” And from a respondent with mental illness,

> “When I go birding by myself, I often keep moving slowly much of the time. When with my mom, we sit down frequently, and taking breaks by sitting in one spot allows me to see birds that may not have revealed themselves if I was just passing by!”

3.3 Styles of birding used

Although a question regarding the style/s of birding used was not asked of respondents, 74 people who responded to the question, above, provided information in their response about one or more of the style/s of birding they engaged with. This data is shared as illustrative of the inherent modifiability of birding. Indeed, as summarized by one respondent, “As I become less mobile, birding is the one thing out in nature I can still do.” Note that the list of styles shared in Figure 3 may not be a complete list of the styles of birding survey respondents, or disabled or nondisabled birders, utilize.

Figure 3: Styles of birding used by disabled birders. Note that 74 people (70%) who responded to the question about positives of birding differently to others shared information about the style/s of birding they engage with. Percentages have been calculated based on the number of people who provided this information.



4. Discussion and Implications

4.1 A strengths-based approach could support inclusion and wellbeing of disabled birders

Disability is often viewed in a negative light (e.g., Mitter et al., 2019), with disabled people frequently stereotyped as dependent, generally incompetent, passive, and to be

avoided (Nario-Redmond, 2010). Results from this study directly contradict these stereotypes as survey respondents identified a wealth of positive experiences as disabled birders. Indeed, the two most common codes related to participants noticing more birds and noticing more details (e.g., of birds' plumage, behavior or habitat). Keeping lists of the number of bird species identified during an outing, in one's backyard, and over one's lifetime is a component of birding enjoyed by many birders (Connell, 2009); noticing more birds and/or more details is certainly a strength when keeping a list. Further, being aware of subtle differences in plumage between species, or between calls or songs, can sometimes be the difference between feeling confused and successfully identifying a bird (Pieplow, 2012). As a result, noticing more details about birds, their behavior, or their habitats could lead to a more enriching and/or satisfying experience for the individual, and for others birding with them. The cultural assumption that disability is a "tragedy" is challenged by this finding.

It is unknown whether respondents to this survey had previously reflected on their access challenge/s as a positive in the context of birding. Although this follow-up question was not asked, it is possible that simply the act of being asked this question in this survey may have begun a positive reframing process for participants and their relationship with their disability through the process of priming (Wang & Dovidio, 2011). One response suggests this may be the case: "It's true that having to sit for a rest or not walk far means there's more time to study what's around you. I hadn't really thought about it, though." Encouraging disabled birders to consider a strengths-based approach to birding could act as a larger-scale prompt for disability and birding organizations, and disabled birders individually, to continue to reflect on and celebrate ways that birding can lean on a disabled birder's strengths. With this in mind, healthcare providers, engagement and program staff and volunteers at nature organizations, and other birders may consider opportunities to amplify and support disabled birders' strengths, and ways to create opportunities for those strengths to be further developed in the context of birding.

The literature reports various strengths that align with our findings. Several studies (e.g. Schippers et al., 2022; Sedgwick et al., 2019) report that for adults with ADHD one of the most common strengths was participants' ability to hyperfocus (see sections 4.2 for more). The participants in Sedgwick et al. (2019) also reported an appreciation for beauty and gratitude, both of which were themes expressed by respondents in this study. Similarly, Nocon et al. (2022) found that people with autism scored highly on appreciation of beauty and gratitude on the VIA Character Strengths Survey (VIA Institute on Character, 2025). Other strengths reported in all of these studies were related to the individual's character strengths (such as kindness, humility), and were not discussed by participants in our study.

Staff and volunteers at nature-related organizations and healthcare practitioners can support a strengths-based approach to birding by designing and implementing bird

outings that utilize various styles of birding. Such outings may include birding by ear events or stationary birding events. One respondent shared, “As I become less mobile, birding is the one thing out in nature I can still do”, which highlights the potential of stationary bird-related programming to be accessible to people with disabilities. Offering different styles of bird outings may emphasize different strengths, and may be more, or less, accessible to different individuals. Indeed, many respondents stated elsewhere in the survey that they enjoyed both of these styles of birding. Additional actions that nature organizations could take to support a strengths-based approach to birding include providing a variety of loaner adaptive equipment, creating events that explicitly provide flexible opportunities for engagement (including shorter events, or come-and-go-as-you-like type events), and mindful birding programs, or programs that support creativity and connect to birds. Doing so may provide participants with the opportunity to discover, nurture and celebrate their strengths in this activity, and this variety of bird outings will likely increase the overall accessibility and inclusiveness of the organizations’ programmatic offerings. With 39% of birders reporting some kind of accessibility challenge (Sinkular et al., 2024), supporting a variety of ways to enjoy birds may increase engagement, regardless of whether individuals identify as “disabled” or not. As Rolim et al. (2021) found, planning for including people with diverse access needs helps facilitate more accessible birding environments for all birders. Doing so could contribute to decreased feelings of self-consciousness among disabled birders as their autonomy and feelings of satisfaction increase with their increased independence in birding (Åberg et al., 2005) – a possible avenue for future research.

4.2 Neurodivergent traits as strengths, rather than deficits, in birding

Despite the typical cultural deficit-based framing of neurodivergences (e.g., encouraging autistic people to mask), neurodivergent birders identified ways their differences enhanced their birding experience. Five of the six neurodivergent birders in this sample reported they noticed more details, had more sensitive hearing which helped them identify more birds, or could use their hyperfocus to learn more about birds. Indeed, Crane et al. (2009) found that autistic adults experience a significant difference in sensory processing compared to neurotypical adults, including in the area of sensory sensitivities. Using a strengths-based approach to birding with a disability shows that something that can be disabling in other environments, such as a sensitivity to sounds, can be a strength while birding.

Exposing people who are neurodivergent to birding may help them develop a special interest in birds, and find that their differences can not only be a strength in this activity, but may be valued by other birders who do not possess similar strengths. This could lead to an increased sense of belonging and social connectedness for neurodivergent birders within the birding community – something that many neurodivergent people lack, to the detriment of their health (Santuzzi & Waltz, 2016;

Stice & Laver, 2019). Future research could explore whether this is the case if such a program was developed. Program directors, environmental educators and bird outing leaders could support the development of birding by ear skills for all birders, and encourage neurodivergent birders to consider sharing these skills and observations with others to create a richer experience for all. It is important that such programs are developed and implemented using neurodiversity-affirming principles, too, including welcoming diverse forms of communication, behavior and self-expression, making accommodations for an individuals' access needs, and not using bribes or withholding a desired outcome to force participation in a certain way (Dallman et al., 2022).

One participant in this study shared a strength that could also act as a weakness: their hyperfocus. Although helping them notice more details, they stated that it might also cause them to miss other birds. (See 3.2.14 Negatives of being a disabled birder.) Hyperfocus is a common trait of people with ADHD and autism, defined as "complete absorption in a task", which often results in improved task performance (Ashinoff & Abu-Akel, 2021).

4.3 Using an intersectional approach in inclusion efforts is vital

Although this survey did not ask demographic questions about gender, race, or ethnicity, six respondents mentioned these aspects of their identity within their answer to this question. Three respondents stated that their primary diagnosis was "being female," with a fourth sharing, "not having a man's bladder", each time in relation to needing access to bathrooms. It is possible that other respondents also highly value access to bathrooms while birding, and a lack of bathrooms may exclude them from visiting a particular birding location, potentially including those with chronic illnesses such as irritable bowel syndrome and those with spinal cord injuries, which can impact bowel and bladder function (Squair et al., 2019). Indeed, Sinkular et al. (2024) found that disabled birders desire more accessible bathrooms at birding locations. Other research on gender and birding (e.g., Cooper and Smith, 2010; Randler and Marx, 2022) indicate that there are differences in how men and women experience birding and the support they receive from others to participate in it. If birding and conservation organizations want to fully include disabled birders, we recommend they work within the social, cultural and institutional environments within the birding and outdoor communities to reduce or eliminate barriers created by sexism, racism, homophobia, transphobia, ageism and other forms of othering and discrimination (Jennings et al., 2024; Soares et al., 2023; Thompson, 2020), while also improving the physical accessibility of birding locations. Additionally, birding and nature organizations could develop and uphold clear guidance on appropriate interpersonal behavior for staff, volunteers, members and participants, including all outing leaders supporting all participants' access needs – which will often likely be different to their own – including access to bathrooms.

4.4 Next steps for organizations wishing to support disabled birders' strengths

These findings may encourage bird and conservation agencies and organizations to apply a strengths-based lens not just to their programs, but to their staff and volunteer recruitment and management as well. Employing disabled wildlife viewers as consultants, which inherently relies on their strengths, can help land and program managers develop initiatives that are relevant and effective for other disabled wildlife viewers, as input from target audiences has been found to be effective in managing wildlife recreation (Manfredo et al., 2002). Nature-based organizations could consider supporting disability affinity groups among their staff, volunteers, members and stakeholders to further this impact, which may be especially important considering that people with disabilities are often socially isolated in their communities (Santuzzi & Waltz, 2016).

Six respondents mentioned using adaptive equipment, indicating that this may be an important consideration in promoting a strengths-based approach to birding. Healthcare providers have long utilized adaptive equipment to enable participation in meaningful activities (e.g. American Occupational Therapy Association, 2020). In birding, adaptive equipment may include mounts to attach optics to wheelchairs, monoculars to retain some peripheral vision, loaner beach or off-road wheelchairs, and bird identification apps, especially for people with learning disabilities, hearing loss or low vision. To support disabled birders, birding groups and nature centers could consider offering loaner adaptive equipment beyond standard binoculars.

Finally, multiple respondents (14%) mentioned "'seeing"' birds, however the word "see" may have been used by many of these respondents figuratively (as in, they both saw and heard birds) instead of literally (they only noticed birds visually), such as, "I *see* lots of great stuff when I'm lost or resting." [Emphasis added.] Research has shown that humans demonstrate a strong bias on relying more on visual information than other sensory input (Colavita, 1974; Sinnett et al, 2007). Were participants who saw more birds only looking at birds in these instances, or were they listening too? Either way, organizations could encourage their bird outing leaders to incorporate auditory birding skills into their programming and hold specific birding by ear events, to both enhance the overall skills of sighted birders, and to utilize a strengths-based approach to birding for birders who have more strengths in auditory identification skills (Mahmoudzadeh & Sarjaz, 2018).

5. Conclusion

Birding is an activity enjoyed by millions of people all over the world (e.g. Randler et al., 2022; eBird, 2025), and many people around the world live with one or more disabilities. As the pervasive cultural view of disability slowly shifts from viewing disability as something tragic and stigmatized that needs to be "fixed", to a more functional and

affirming model of disability, using a strengths-based approach celebrates an individual's differences without harmful messaging that they are somehow “defective”.

Many disabled birders report positive birding experiences as a result of, or in spite of, their disability. The variety and richness of the strengths identified highlights the adaptability and versatility of birding, and the breadth of disability and human interests. Being able to adapt one’s birding style supports birders – disabled or not – in participating in this activity regardless of their energy levels, pain, disability, available time or other activity limitations, and is an affordance that very few hobbies offer. Being able to access such positives through a free, or very low cost, activity may be particularly important when mental healthcare is unavailable or unaffordable, as is the case for many people with disabilities around the world (Gréaux et al., 2023). Indeed, an increase in bird diversity has been shown to correlate with a decrease in hospitalizations for anxiety and mood disorders, such as Major Depressive Disorder, in the general population (Buxton et al., 2023), indicating the potential for birding to be used as a therapeutic modality for people living with mental illness.

Promoting a strengths-based approach to birding with a disability could help counter the negative impacts of disability stigma, which can improve health and social outcomes for up to a quarter of the population, further the de-stigmatization of disability, and increase engagement of disabled people in birding and conservation efforts. Doing this creates an opportunity to invite a significant proportion of the population into conservation efforts (Cooper et al., 2015), creating a more accessible and equitable world – for the benefit of birds *and* humans.

Declaration of generative AI and AI-assisted technologies in the writing process

No generative AI or AI-assisted technologies were used during the writing process.

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Declaration of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

Freya McGregor conceptualized this project, curated the data, conducted formal analysis and methodology, and completed writing – original draft.
Ashley Dayer conducted funding acquisition, supervision and writing – review and editing.
Emily Sinkular conducted formal analysis and writing – review and editing.

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Supplementary Material: Glossary of key words or phrases used in this paper

Word or phrase	*Explanation*
Ableism	Negative biases, stereotypes, and discrimination against people with disabilities, often in the form of assumptions that people with disabilities are less capable, and only nondisabled people will want to participate in an activity or visit a place, etc (National Center on Disability and Journalism, 2021).
Access needs	Features of a program or environment that someone requires in order to be able to participate in or use it (Reinholz & Ridgway, 2021). Rather than focusing on a person's medical history (such as, they have a C5-C6 incomplete spinal cord injury), shifting the emphasis to their access needs is more respectful, relevant and functional, as it gives us information about what we might need to change to allow them to participate (e.g. this person likely uses a power wheelchair, so their access needs may include no steps and a firm and stable trail surface).
Accessibility challenge	"The difficulties someone experiences in interacting with or while using the physical or social environment while trying to engage in a meaningful activity (such as wildlife viewing). This may be a result of a mobility challenge, blindness or low vision, intellectual or developmental disabilities (including Autism), mental illness, being Deaf or Hard of Hearing, or other health concerns" (Sinkular et al., 2024).

Birder/birding	The nonconsumptive act of enjoying wild birds, which may involve appreciating birds by sight, sound, or otherwise noticing their presence (McGregor, 2021). “Birding” and “birder”, rather than “birdwatching” or “birdwatcher”, have been used throughout this paper, as many people enjoy birds by sound, too, and not just visually.
Disabled birder / birder with a disability	Many people in the disability community prefer identity-first language (e.g., “disabled person”, “Deaf person”, “autistic person”), while others prefer person-first language (e.g., “person with a disability”, “person with cerebral palsy”, “person who uses a wheelchair”) (National Center on Disability and Journalism, 2021). In an attempt to honor the nuances of this broad community, these two phrases are used interchangeably throughout this paper.
Disability	“A physical or mental impairment that substantially limits one or more of the major life activities of [an] individual” (Americans with Disabilities Act, Title 42, §12102). “Physical or mental impairment means: (i) Any physiological disorder or condition, cosmetic disfigurement, or anatomical loss affecting one or more body systems, such as: neurological, musculoskeletal, special sense organs, respiratory (including speech organs), cardiovascular, reproductive, digestive, genitourinary, immune, circulatory, hemic, lymphatic, skin, and endocrine; or (ii) Any mental or psychological disorder such as intellectual disability, organic brain syndrome, emotional or mental illness, and specific learning disability” (Americans with Disabilities Act, Title 42, §12102). “Major life activities include, but are not limited to: Caring for oneself, performing manual tasks, seeing, hearing, eating, sleeping, walking, standing, sitting, reaching, lifting, bending, speaking, breathing, learning, reading, concentrating, thinking, writing, communicating, interacting with others, and working” (Americans with Disabilities Act, Title 42, §12102).
Deaf, deaf and hard of hearing	Deaf (with a capital “D”) refers to members of the Deaf community who primarily use sign language to communicate, rather than speech and hearing, and celebrates the culture surrounding this

	community. Lowercase “deaf” refers to people with hearing loss, however some people prefer identifying as “hard of hearing” (National Center on Disability and Journalism, 2021). Using all three terms at once allows us to include all of these different identities.
Neurodivergent	A community-generated term and identity, not a medical diagnosis, created to celebrate and affirm the various ways peoples’ brains can operate differently from the “typical”. Includes people who are autistic, who have attention deficit hyperactivity disorder (ADHD), dyslexia, dyscalculia, and other conditions, whether formally diagnosed or self- or peer-diagnosed. Depending on the definition, may also include people living with depression, anxiety, bipolar disorder, post traumatic stress disorder (PTSD), schizophrenia and other formal mental health diagnoses (Nerenberg, 2021).